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Bitcoin's Limited Adoption Problem

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1 Big picture

- **Question.** Can Bitcoin's economic design support widespread adoption as a decentralized payment system, or does its design itself generate limited adoption as an equilibrium outcome?
- **Central thesis.** Bitcoin's limited adoption is not merely a transitional feature of a young technology. In the model, limited adoption is an equilibrium implication of consensus, network delay, and endogenous mining entry.
- **Key mechanism.** As transaction demand rises, congestion raises transaction fees. Higher fees attract more miners. A larger mining network increases network delay. Network delay raises the probability of forks and prolongs the time needed to regain consensus. Longer settlement times reduce user utility and push users to an outside payment option.
- **Why Bitcoin differs from centralized payments.** A centralized payment system can usually respond to high demand by increasing processing capacity. Bitcoin cannot do this costlessly because a higher transaction rate also raises fork probabilities and consensus delays.
- **Main result 1.** With a fixed transaction rate, Bitcoin faces limited adoption: as transaction demand becomes large, the equilibrium adoption rate converges to zero.
- **Main result 2.** Even if Bitcoin is allowed to increase its transaction rate with demand, limited adoption persists because consensus wait time diverges when transaction rates become very large.
- **Finite-demand result.** The paper also shows that for any finite transaction demand, Bitcoin faces a finite upper bound on total adoption. The bound is far below the scale of traditional payment systems under reasonable parameter values.
- **Practical implication.** Matching a traditional system such as VISA would require transaction rates so high that almost every block would fork, and settlement times would become prohibitively long.
- **Potential remedies.** Partial centralization of mining can improve adoption by lowering network delay, but it does not eliminate the adoption bound. Designs that bound fork probabilities away from one, such as permissioned blockchains or other protocol changes, can avoid the limited adoption problem.
- **Economic takeaway.** Bitcoin has negative network effects. More potential users can induce more miners and more congestion, but consensus delay grows with network size and undermines adoption.

2 Model setup and primitives

2.1 Users, transaction demand, and outside option

- There are N potential users, each with one transaction that can be submitted to Bitcoin.
- Each user has a type $c_i \in [\underline{c}, \bar{c}]$ that measures disutility from waiting for settlement.
- User types are ordered by impatience: a larger c_i means greater waiting cost.
- Users can choose whether to adopt Bitcoin or use a traditional alternative payment system.
- The traditional alternative is normalized to yield zero utility.
- A Bitcoin user receives a network utility benefit from adoption, but pays a transaction fee and suffers waiting costs.

The adoption cutoff is denoted by c_* , and the equilibrium adoption rate is

$$\pi_* = \Pr(c_i \leq c_*) = \frac{c_* - \underline{c}}{\bar{c} - \underline{c}}. \quad (1)$$

Equivalently,

$$c_* = (\bar{c} - \underline{c})\pi_* + \underline{c}. \quad (2)$$

Interpretation: The cutoff formulation means that low waiting-cost users are more likely to adopt Bitcoin, while high waiting-cost users opt out. This is the standard adverse-selection-style adoption margin in the model.

2.2 Utility from Bitcoin

The utility component from using Bitcoin is

$$R_1(\pi_* N)^\alpha + R_0, \quad (3)$$

where:

- R_0 is a baseline utility from Bitcoin;
- $R_1(\pi_* N)^\alpha$ is the benefit from the size of the adopting network;
- $\alpha \in (0, 1)$ governs the strength of positive user-side network effects.

If user i pays fee f_i and waits W_i , her Bitcoin utility is approximately

$$R_1(\pi_* N)^\alpha + R_0 - c_i \mathbb{E}[W_i] - f_i. \quad (4)$$

Interpretation: The model allows Bitcoin to have positive adoption externalities through $R_1(\pi_* N)^\alpha$. The main contribution is to show that Bitcoin's consensus mechanics can overwhelm these positive network effects.

2.3 Miners and free entry

- Miners store copies of the ledger and compete to update the blockchain.
- A miner pays cost $\beta > 0$ to acquire mining technology.
- Miners receive block rewards B and transaction fees.
- The analysis focuses on transaction fees, while allowing a block reward.
- Mining has free entry: miners enter until expected mining profit equals the cost β .

If there are M miners, each miner's expected payoff from mining is

$$\frac{1}{M} \left(B + \mathbb{E} \sum_i f_i \right) - \beta. \quad (5)$$

The free-entry condition is therefore

$$M = \frac{B + \mathbb{E} \sum_i f_i}{\beta}. \quad (6)$$

Interpretation: Transaction fees do not just compensate miners. They also endogenously determine the size of the mining network. This is crucial because more miners increase network delay.

2.4 Blockchain, forks, and consensus

- The blockchain is a ledger updated by blocks.
- A block is valid if it solves the computational puzzle.
- A valid block may still fail to enter the consensus chain if another inconsistent block is produced before all miners become aware of the first block.
- Network delay is the time required for information to propagate through the mining network.
- A fork occurs when miners disagree on which valid block should be appended to the chain.

The fork probability is

$$\Pr(Fork) = 1 - e^{-\Lambda \Delta(M)}, \quad (7)$$

where Λ is the transaction/block processing rate and $\Delta(M)$ is average network delay for a normalized block across the mining network.

Interpretation: Fork probability increases in both transaction rate and network delay. Higher transaction rates make blocks arrive closer together. Higher network delay means that miners remain unaware of one another's blocks for longer. Both raise the chance of disagreement.

2.5 Network delay assumptions

The network delay function satisfies:

$$\Delta(1) = 0, \quad \Delta'(M) > 0 \quad \text{for } M > 1. \quad (8)$$

- With one miner, there is no communication delay.
- With more than one miner, communication delay is positive and increasing.
- The paper keeps the function general, so the results are not tied to a special network topology.

Interpretation: This is the minimum structure needed to capture decentralization. Decentralization means many miners, and many miners mean communication delay.

3 Equilibrium definition

3.1 Objects in equilibrium

A Bitcoin equilibrium consists of:

- an adoption cutoff c_* ;

- a fee function $\phi(c)$ mapping each user type to a transaction fee;
- a set of fee realizations for all users;
- an equilibrium number of miners M .

Interpretation: The equilibrium is not just a user-adoption equilibrium. It simultaneously determines fees and mining network size, and those variables feed back into wait times and adoption.

3.2 User adoption condition

A user adopts Bitcoin if and only if using Bitcoin is optimal:

$$c_i \leq c_* \iff \max_{f \geq 0} \{R_1(\pi_* N)^\alpha + R_0 - c_i \mathbb{E}[W(f, f_{-i})] - f\} \geq 0. \quad (9)$$

Interpretation: The cutoff user is indifferent between Bitcoin and the traditional payment system. Users with lower waiting disutility adopt; users with higher waiting disutility do not.

3.3 Fee function

The equilibrium fee function solves

$$\phi(c_i) = \arg \max_{f \geq 0} \{R_1(\pi_* N)^\alpha + R_0 - c_i \mathbb{E}[W(f, \phi(c_{-i}))] - f\}. \quad (10)$$

The paper shows that for adopting users,

$$\phi(c_i) = \frac{N-1}{\bar{c} - \underline{c}} \times \frac{c_i^2 - \underline{c}^2}{2\Lambda}, \quad (11)$$

while non-adopting users pay zero fees.

Interpretation: The fee increases in a user's waiting disutility c_i . More impatient users pay more because a reduction in waiting time is more valuable to them. Fees also rise with transaction demand N and fall with the transaction rate Λ .

3.4 Wait time and priority service

Transactions are processed in descending fee order. A user who pays a lower fee must wait behind users who pay higher fees. The user's wait time is

$$W(f_i, f_{-i}) = \sum_{j: f_i \leq f_j} H_j + Z_i, \quad (12)$$

where H_j is the service time for transaction j and Z_i is the time until the blockchain next reaches consensus.

Interpretation: The expected wait time has two components: a conventional queuing component and a consensus component. The latter is specific to Bitcoin and other decentralized blockchains.

3.5 Proposition 1: equilibrium characterization

The equilibrium is characterized by:

- the equilibrium fee function;
- the free-entry number of miners;
- an adoption cutoff condition;
- a full-adoption condition for cases in which all users adopt.

Interpretation: Proposition 1 is the engine of the paper. It links user fees, miner entry, adoption, and consensus wait times in a single equilibrium system.

4 Main results: why Bitcoin has limited adoption

4.1 Proposition 2: fixed transaction rate

If Bitcoin keeps its transaction rate fixed, then Bitcoin faces limited adoption:

$$\lim_{N \rightarrow \infty} \pi_* = 0. \quad (13)$$

Mechanism

1. Higher transaction demand increases congestion.
2. Congestion raises equilibrium transaction fees.
3. Higher transaction fees attract more miners through free entry.
4. More miners raise network delay.
5. Higher network delay raises consensus wait times.
6. Higher wait times reduce the value of using Bitcoin.
7. The adoption cutoff falls relative to the size of potential demand.

Interpretation: Bitcoin cannot maintain a non-negligible adoption rate as demand grows. The problem is not only limited block capacity. It is that fees, mining entry, and network delay interact to make settlement increasingly unattractive.

4.2 Proposition 3: variable transaction rate

A natural solution is to let the transaction rate Λ_N rise with transaction demand N . Proposition 3 shows that even this does not eliminate limited adoption:

$$\lim_{N \rightarrow \infty} \pi_* = 0. \quad (14)$$

Interpretation: Increasing processing capacity works in centralized systems, but it fails in Bitcoin because the consensus component of wait time grows when high transaction rates create frequent forks.

4.3 Wait-time decomposition

For user i , the expected wait time can be decomposed as

$$\mathbb{E}_i[W(f_i, f_{-i})] = \underbrace{\frac{(N-1)(c_* - c_i) + 1}{\Lambda_N}}_{\text{traditional wait time}} + \underbrace{\tau(\Lambda_N, \Delta(M))}_{\text{consensus wait time}}. \quad (15)$$

- The first term is the traditional queuing wait time.
- The second term is the consensus wait time.
- A centralized payment system has only the first term.
- Bitcoin has both terms.

Interpretation: The traditional wait time can be made small by increasing Λ_N . The consensus wait time cannot be made small in the same way because higher Λ_N raises fork probabilities.

4.4 Proposition 4: consensus wait time diverges

If Bitcoin increases transaction rates to keep up with transaction demand,

$$\lim_{N \rightarrow \infty} \Lambda_N = \infty, \quad (16)$$

then

$$\lim_{N \rightarrow \infty} \tau(\Lambda_N, \Delta(M)) = \infty. \quad (17)$$

Interpretation: Increasing the transaction rate reduces conventional queuing time but raises fork probability. When fork probability becomes high, the system spends too much time trying to regain consensus.

4.5 Proposition 5: no adoption problem without network delay

If network delay is counterfactually zero for all network sizes, then widespread adoption can be obtained even with arbitrarily large networks.

Interpretation: This proposition is important because it isolates the source of the problem. Limited adoption is not a generic property of any transaction system; it arises from the combination of consensus and network delay.

5 Finite transaction demand and practical adoption limits

5.1 Why finite- N results matter

The limiting results establish that Bitcoin cannot scale to arbitrary demand. The paper then asks whether the same logic has practical force for realistic transaction volumes.

Interpretation: This section bridges theory and practice. It shows that the adoption problem is not merely an asymptotic curiosity.

5.2 Proposition 6: finite bound on adoption level

With a fixed transaction rate, there is a finite bound on total adoption:

$$\sup_{N \in \mathbb{N}} \pi_* N < \infty. \quad (18)$$

The key inequality is

$$R_1 \left(\frac{c_* - \underline{c}}{\bar{c} - \underline{c}} \right)^\alpha N^\alpha + R_0 \geq \frac{N - 1}{\bar{c} - \underline{c}} \times \frac{c_*^2 - \underline{c}^2}{2\Lambda}. \quad (19)$$

Interpretation: The left side is the benefit from Bitcoin. The right side is the marginal user's fee. If total adoption becomes arbitrarily large, fees grow too large to be consistent with adoption.

5.3 Proposition 7: finite bound with variable transaction rate

Even if the transaction rate varies with demand, Bitcoin still faces a finite bound on adoption:

$$\sup_{N \in \mathbb{N}} \pi_* N < \infty. \quad (20)$$

The key logic is that either the transaction rate grows too slowly, causing fees to be too high, or it grows fast enough to avoid high fees, causing consensus wait time to become too high.

Interpretation: Bitcoin faces a dilemma. Low transaction rates cause congestion; high transaction rates cause forks and consensus delays.

5.4 Proposition 8: practical limits of adoption

The paper benchmarks Bitcoin against the transaction volume of VISA in the United States: approximately 150 million transactions per day.

If Bitcoin attempted to process such a volume, then:

- each block would correspond to a fork with over 99% probability;
- the blockchain would be in a state of a fork with over 99% probability;
- the expected time for transaction settlement would become prohibitive;
- under the paper’s calculations, expected settlement time would exceed one year.

Interpretation: The finite- N results imply that Bitcoin is unlikely to replace high-volume centralized payment systems under its current consensus architecture.

6 Overcoming limited adoption

6.1 Partial centralization of mining markets

Proposition 9 compares two mining networks:

- a relatively decentralized network with delay function $\Delta_D(M)$;
- a relatively centralized network with delay function $\Delta_C(M)$;
- where $\Delta_C(M) \leq \Delta_D(M)$ for all M .

The result is:

$$\pi_*^C \geq \pi_*^D. \quad (21)$$

Interpretation: Partial centralization raises adoption because it lowers network delay and fork probability. This explains why mining pools and centralized coordination can improve performance. But the result does not say centralization is costless: it weakens the decentralization property that motivates Bitcoin.

6.2 Beyond Bitcoin

The paper then studies blockchain designs in which the fork probability is not Bitcoin’s specific function

$$1 - e^{-\Lambda \Delta(M)}. \quad (22)$$

Instead, suppose

$$\Pr(\text{Fork}) = p_F(\Lambda, \Delta(M)). \quad (23)$$

6.3 Proposition 10: overcoming limited adoption

If the fork probability is bounded away from unity,

$$\sup_{\Lambda, \Delta} p_F(\Lambda, \Delta) < 1, \quad (24)$$

then widespread adoption can be obtained in equilibrium even with an arbitrarily large network.

Interpretation: The problem is not decentralization alone. The core issue is that Bitcoin’s design lets fork probability approach one when the transaction rate becomes too high. Designs that prevent this can support more adoption.

6.4 Protocol examples

- **Expedited transaction rates.** Simply increasing block sizes or block rates does not solve the problem because it raises fork probability.
- **Bitcoin NG.** A miner is authorized for an epoch to append many transactions, reducing fork risk within that epoch.
- **Modern proof-of-stake protocols.** Some protocols reduce consensus delays or bound fork probabilities through validator selection and finality mechanisms.
- **Permissioned blockchains.** A known and limited set of validators can coordinate faster, reducing delay and the probability of forks.

Interpretation: The paper’s results are not anti-blockchain in general. They are a critique of Bitcoin’s specific consensus and scaling architecture.

7 Identification, empirical interpretation, and limitations

7.1 What the paper identifies theoretically

- The paper identifies a structural mechanism linking adoption, fees, miner entry, network delay, forks, and consensus.
- It provides mathematical conditions under which limited adoption arises as an equilibrium property.
- The results are not based on empirical estimation but on equilibrium comparative statics.

Interpretation: The contribution is conceptual and theoretical. It explains why Bitcoin may remain limited even if demand grows and users value the network.

7.2 Why the model matters for fintech and blockchain design

- The model clarifies that decentralized consensus is not merely a technical issue; it is an economic constraint.
- The fee market, mining entry, and settlement delay jointly determine adoption.
- High fees can attract miners, but miner entry worsens the delay problem.
- Scaling proposals must address consensus, not just transaction throughput.

Interpretation: A payment system's scalability cannot be judged only by raw transaction capacity. It must also be judged by how transaction capacity interacts with consensus and settlement finality.

7.3 Limitations of the model

- Users have simple one-unit transaction demand.
- The model abstracts from speculative motives for holding Bitcoin.
- The model focuses on payments, not Bitcoin's role as a store of value or investment asset.
- Network delay is modeled in general form rather than estimated from detailed network topology.
- Mining is represented through free entry and fee/reward tradeoffs, abstracting from strategic mining behavior beyond forks.
- The outside payment option is normalized rather than modeled explicitly.

Interpretation: The abstraction is appropriate for isolating the adoption mechanism, but the model does not claim to explain all sources of Bitcoin value or all blockchain adoption patterns.

8 Original equation and proposition panels

The original paper has no empirical tables or plotted figures. To preserve the direct-extraction style, this section uses compact cropped panels of the main original equations, definitions, and propositions.

8.1 Model primitives: adoption, miners, and fork probability

Read:

- The first panel shows the adoption cutoff, miner payoff, and free-entry mining condition.
- The second panel shows the fork probability, which is the central technological channel in the model.
- Fork probability is increasing in the transaction rate and network delay.

Economic message: These primitives create the feedback loop: higher fees bring more miners; more miners raise delay; delay raises fork risk; fork risk lengthens consensus.

to as the marginal user hereafter. We define the adoption rate, π_* , as the expected proportion of users that adopt Bitcoin (i.e., $\pi_* \equiv \mathbb{P}(c_i \leq c_*)$). Consequently, the cut-off, c_* , possesses a bijective relationship with the adoption rate, π_* , as follows:

$$\pi_* = \frac{c_* - \underline{c}}{\bar{c} - \underline{c}}, \quad c_* = (\bar{c} - \underline{c}) \cdot \pi_* + \underline{c} \quad (2)$$

2.2. Miners

A miner refers to a member of the Bitcoin network that stores a copy of the Bitcoin ledger and “mines” to update the ledger. As a convention, we use “miner” to refer to a single computer processor that participates in the Bitcoin network. This convention does not preclude that multiple processors might be controlled by the same agent. In fact, we discuss the implications of such centralization in [Section 5.1](#) although our main results hold for arbitrary mining market structures.

Each miner must pay some cost $\beta > 0$ to acquire mining technology. We assume that each processor possesses identical hashing power so that each miner expects to earn an equal share of fees among all miners.² Our analysis focuses on transaction fees, but we allow that there exists a block reward, B , paid to miners.³ Then, assuming risk neutral preferences, each miner solves:

$$\max\left\{\frac{1}{M}\left(B + \mathbb{E}\left[\sum_i f_i\right]\right) - \beta, 0\right\} \quad (3)$$

M denotes the equilibrium number of miners so that the pay-off from mining equals $\frac{1}{M}\left(B + \mathbb{E}\left[\sum_i f_i\right]\right)$ whereas the pay-off from not mining equals 0. Bitcoin's design allows for free entry among miners, so miners must be indifferent between mining and not mining. Accordingly, we impose the following free entry condition in equilibrium:

$$M = \frac{B + \mathbb{E}\left[\sum_i f_i\right]}{\beta} \quad (4)$$

As an aside, our model assumes that miners are patient.

Adoption cutoff and miner free entry

Miner j first receives news of Miner i 's block. If any Miner $j \neq i$ finds that Miner i 's block is not consistent with her blockchain when she first receives news of Miner i 's block then she rejects the block and a fork arises due to a disagreement between Miners i and j . Under such an assumption, if all miners initially store the same blockchain (i.e., the blockchain is in a state of consensus) then Miner i 's block will not correspond to a fork if no Miner $j \neq i$ produces a valid block before receiving news of Miner i 's block. This fact holds because two blocks are necessarily inconsistent with each other if they are produced without knowledge of each other.⁸ Intuitively, if Miner $j \neq i$ produces a valid block after Miner i but before receiving news of Miner i 's block then Miner i and j disagree on the temporal order of their two blocks because each miner believes that her block was produced first. As the two blocks are inconsistent with each other, each miner must select only one of the two blocks. Since either a time-ordering rule or even personal self-interest would lead to each miner selecting her own block, a fork results (i.e., lack of consensus obtains).

To incorporate forks into our model, we proceed by determining the probability that a given valid block corresponds to a fork:

$$\mathbb{P}(\text{Fork}) = 1 - e^{-\Lambda \Delta(M)} \quad (5)$$

Λ denotes the transaction rate, and $\Delta(M)$ denotes the network delay for a normalized single-transaction block averaged over all pairs of the M miners in the network. [Equation \(5\)](#) highlights that the fork probability increases in both the transaction rate and the network delay for normalized block; we establish that relationship in

Fork probability and network delay

8.2 Equilibrium and fee function

Read:

- Definition 1 formally lists the equilibrium objects: adoption cutoff, fee function, fees, and mining network size.
- Proposition 1 characterizes equilibrium fees, miner entry, and the adoption cutoff.
- The equilibrium fee function increases with user impatience and transaction demand.

Economic message: The equilibrium is a joint fixed point in adoption, transaction fees, and miner entry. This joint determination is what creates negative network effects.

2.4. Equilibrium

We formally define an equilibrium as follows:

Definition 1. Bitcoin Equilibrium

Our Bitcoin model is parameterized by the number of users, N , three parameters relating to the blockchain utility for each user, R_0 , R_1 and α , and the blockchain transaction rate, Λ . The N users possess types, $\{c_i\}_{i=1}^N$, with $c_i \sim U[\underline{c}, \bar{c}]$. A Bitcoin Equilibrium is (1) an adoption cut-off, c_* , (2) a function, $\phi(c)$, that maps user types to their fees, (3) a set of fee realizations for the N users, $\{f_i\}_{i=1}^N$, and (4) a mining network size, M . The equilibrium satisfies the following conditions:

- (i) A User Adopts Bitcoin Iff Doing So Is Optimal A user adopts Bitcoin (i.e., $c_i \leq c_*$) if and only if doing so is optimal.

$$\text{(i.e., for all } c \leq c_* \Leftrightarrow \max_{f \geq 0} R_1 \cdot (\pi_* N)^\alpha + R_0 - c \cdot \mathbb{E}[W(f, f_{-i})] - f \geq 0 \text{ where } \pi_* = \frac{c_* - \underline{c}}{\bar{c} - \underline{c}})$$

- (ii) Equilibrium User Fees Depend Upon User Types User i pays a fee, f_i , which depends on her type, c_i . (i.e., for all i : $f_i = \phi(c_i)$)

- (iii) Equilibrium Fee Function Provides Optimal Bitcoin Fees

The fee function, $\phi(c_i)$, provides optimal Bitcoin fees. In particular, the user maximizes her utility from using Bitcoin if she adopts Bitcoin (i.e., $\phi(c)$ solves $\max_{f \geq 0} R_1 \cdot (\pi_* N)^\alpha + R_0 - c \cdot \mathbb{E}[W(f, f_{-i})] - f$ if $c \leq c_*$) but pays zero fees to Bitcoin otherwise (i.e., $\phi(c) = 0$ if $c > c_*$) because, per Condition i, using Bitcoin is not optimal in that case (i.e., $\max_{f \geq 0} R_1 \cdot (\pi_* N)^\alpha + R_0 - c \cdot \mathbb{E}[W(f, f_{-i})] - f < 0$ if $c > c_*$).

- (iv) A User Receives Settlement After Higher Priority Users And When Consensus Next Obtains

User i 's wait time, $W(f, f_{-i})$, if she pays fee f depends not only on f but also on other user fees, f_{-i} . More precisely, transactions are processed in descending fee order so that User i must wait $\sum_{j: f \leq f_j} H_j$ with H_j denoting the service time for User j and $\{j: f \leq f_j\}$ denoting the set of users paying higher fees.⁹ Further, User i 's transaction is not settled even after her transaction enters the blockchain unless the blockchain is at consensus. We denote the (random and possibly zero) time until consensus next obtains by Z_i .¹⁰ Thus, our user wait function becomes:

$$W(f, f_{-i}) = \sum_{j: f \leq f_j} H_j + Z_i$$

Definition 1: Bitcoin equilibrium

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- (v) The Mining Market Is Characterized By Free Entry The total expected cost of mining, β , equals the total expected profit from mining, $\frac{1}{M} \left(B + \mathbb{E}[\sum_i f_i] \right)$, for each miner. (i.e., $\beta = \frac{1}{M} \left(B + \mathbb{E}[\sum_i f_i] \right)$)

Definition 1 simply summarizes the discussion from Sections 2.1–2.3, condensing it into an equilibrium definition. Therefore, we do not elaborate further on our equilibrium definition except that we note now that we assume the blockchain's stationary distribution characterizes its initial state.¹¹ The subsequent result establishes equilibrium existence:

Proposition 1. Bitcoin Equilibrium

There exists a Bitcoin Equilibrium. The following conditions characterize that equilibrium with $\tau(\Lambda, \Delta(M)) \equiv \mathbb{E}[Z_1]$, $\Psi(\Lambda, \Delta(M)) \equiv \frac{1}{\Lambda} + \tau(\Lambda, \Delta(M))$ and $M(c) \equiv \frac{\beta}{B} + \frac{N(N-1)}{6\beta\Lambda} \frac{(c-\underline{c}) \cdot (c^2 + c\underline{c} - 2\underline{c}^2)}{(\bar{c}-\underline{c})^2}$:

- (A) $\phi(c) = \frac{N-1}{\bar{c}-\underline{c}} \times \frac{c^2 - \underline{c}^2}{2\Lambda}$ if $c \leq c_*$ and $\phi(c) = 0$ otherwise.
- (B) $M = M(c_*)$.
- (C) If $R_1 \cdot N^\alpha + R_0 < \bar{c} \cdot \Psi(\Lambda, \Delta(M(\bar{c}))) + (N-1) \times \frac{\bar{c} + \underline{c}}{2\Lambda}$, then $c_* \in (\underline{c}, \bar{c})$ and c_* solves:

$$R_1 \cdot \left(\frac{c_* - \underline{c}}{\bar{c} - \underline{c}} \right)^\alpha \cdot N^\alpha + R_0 = c_* \Psi(\Lambda, \Delta(M(c_*))) + \frac{N-1}{\bar{c} - \underline{c}} \times \frac{c_*^2 - \underline{c}^2}{2\Lambda}$$

Proposition 1: equilibrium characterization

We follow prior literature and require that ϕ constitutes a strictly increasing and twice differentiable function on the interval $(\underline{c}, c_*)$. In such a case, any equilibrium must satisfy the above conditions, and we prove that at least one such equilibrium exists for any set of parameters.

Proposition 1 (A) characterizes the equilibrium fee function. This fee function is such that each user's optimal fee equals the function evaluated at the user's type (i.e., $f_i = \phi(c_i)$) when each user anticipates that all other users select fees according to the same fee function (i.e., for all $j \neq i$: $f_j = \phi(c_j)$). More formally, as required by **Definitions 1** (ii) and (iii), ϕ is a function that solves the following problem:

$$\phi(c_i) = \arg \max_{f \geq 0} R_1 \cdot (\pi_* N)^\alpha + R_0 - c_i \cdot \mathbb{E}[W(f, \phi(c_{-i}))] - f \quad (6)$$

for all c_i with $\phi(c_{-i}) \equiv \times_{j:j \neq i} \phi(c_j)$ so that $\phi(c_{-i})$ corresponds to the equilibrium fee of all users other than User

$$\frac{c_*^2 - \bar{c}^2}{2\Lambda}.$$

Proposition 1 (D) characterizes the adoption cut-off in the case that all users weakly prefer transacting via the blockchain. In this case, $c_* = \bar{c} \Leftrightarrow \pi_* = 1$ because then any User c_i is below the adoption cut-off (i.e., $c_i \leq c_* = \bar{c}$ for all i). Note that this case arises only when the exogenous utility from transacting via the blockchain with full adoption, $R_1 \cdot (\pi_* N)^\alpha + R_0 = R_1 \cdot N^\alpha + R_0$, is sufficiently high such that even the highest-wait-disutility user (i.e., any User c_i such that $c_i = \bar{c}$) weakly prefers to use the blockchain instead of the traditional alternative (i.e., $R_1 \cdot N^\alpha + R_0 \geq \bar{c} \cdot \Psi(\Lambda, \Delta(M(\bar{c}))) + (N-1) \times \frac{\bar{c} + \bar{c}}{2\Lambda}$).

3. Main results

We begin by considering Bitcoin with a fixed transaction rate (i.e., fixed Λ), which is consistent with Bitcoin's existing design. In that context, we demonstrate that Bitcoin faces limited adoption (**Proposition 2**). We then allow for a variable transaction rate within Bitcoin (i.e., we let

8.3 Limited adoption propositions

Read:

- Proposition 2 establishes limited adoption under a fixed transaction rate.
- Proposition 3 shows that limited adoption still arises when transaction rates vary with demand.
- The wait-time decomposition separates the traditional queuing component from the consensus component.

Economic message: Increasing throughput solves the traditional queuing problem but worsens consensus delays when forks become frequent.

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Proposition 3. Limited Adoption II

Suppose that Bitcoin's transaction rate, Λ_N , varies with transaction demand, N . Then, even in such a case, Bitcoin faces limited adoption (i.e., $\lim_{N \rightarrow \infty} \pi_* = 0$).

Proposition 3 allows that Bitcoin's transaction rate, Λ , depends upon transaction demand, N , and we hereafter acknowledge that dependence by denoting the transaction rate as Λ_N . Even allowing for such dependence, **Proposition 3** establishes that Bitcoin faces limited adoption.

To understand **Proposition 3**, we highlight that our model implies the following decomposition for User i 's expected equilibrium wait time, $\mathbb{E}_i[W(f_i, f_{-i})]$, with $\mathbb{E}_i[\cdot]$ denoting an expectation with respect to User i 's information set:

$$\mathbb{E}_i[W(f_i, f_{-i})] = \underbrace{\frac{(N-1)(c_* - c_i) + 1}{\Lambda_N}}_{\text{Traditional Wait Time}} + \underbrace{\tau(\Lambda_N, \Delta(M))}_{\text{Consensus Wait Time}} \quad (8)$$

tion rate (i.e., fixed Λ), which is consistent with Bitcoin's existing design. In that context, we demonstrate that Bitcoin faces limited adoption (**Proposition 2**). We then allow for a variable transaction rate within Bitcoin (i.e., we let Λ vary), and we demonstrate that our conclusion regarding Bitcoin facing limited adoption holds in that case also (**Proposition 3**).

Proposition 2. Limited Adoption I

Bitcoin faces limited adoption (i.e., $\lim_{N \rightarrow \infty} \pi_* = 0$).

Proposition 2: limited adoption with fixed transaction rate

Proposition 3 and wait-time decomposition

8.4 Network delay and finite adoption bounds

Read:

- Proposition 4 shows that consensus wait time diverges with sufficiently high transaction rates.
- Proposition 5 shows that the adoption problem disappears if network delay is counterfactually zero.

- Proposition 6 gives a finite adoption bound for fixed transaction rates.
- Proposition 7 extends the finite bound to variable transaction rates.

Economic message: Network delay is the bottleneck. Bitcoin cannot escape limited adoption by simply processing faster, because faster processing creates forks.

subsequently, the consensus wait time, $\tau(\Lambda_N, \Delta(M))$, becomes prohibitive as the transaction rate increases:

Proposition 4. Consensus Wait Time Diverges with Transaction Demand

Suppose that Bitcoin adjusts its transaction rate to keep pace with transaction demand (i.e., $\lim_{N \rightarrow \infty} \Lambda_N = \infty$). Then, the Consensus Wait Time, $\tau(\Lambda_N, \Delta(M))$, diverges as demand diverges, (i.e., $\lim_{N \rightarrow \infty} \tau(\Lambda_N, \Delta(M)) = \infty$).

To provide some intuition regarding Proposition 4, we return to Eq. (5) which establishes that the probability a block corresponds to a fork equals $1 - e^{-\Lambda \Delta(M)}$.¹² Accordingly, a higher transaction rate, Λ , implies a higher fork probability and thus a lower probability of consensus. In turn, a lower probability of consensus on a given block implies a longer consensus wait time. In fact, Proposition 4 highlights that arbitrarily large transaction rates generate arbitrarily large consensus wait times. Thus, expected wait times diverge as transaction rates become arbitrarily large (i.e., $\lim_{N \rightarrow \infty} \mathbb{E}_i[W(f_i, f_{-i})] = \infty$ when $\lim_N \Lambda_N = \infty$ as per Eq. (8)) so that limited adoption arises even when transaction rates keep pace with transaction demand per Proposition 3.

The implication of our results as a whole is that Bitcoin faces a dilemma in the face of heightened transaction demand. Either Bitcoin keeps transaction rates fixed or it varies transaction rates with transaction demand. If Bitcoin keeps transaction rates fixed (in line with Bitcoin's current design) then Bitcoin faces prohibitive traditional wait times and, in turn, limited adoption per Proposition 2. Alternatively, if Bitcoin varies transaction rates with transaction demand then the consensus wait time becomes prohibitive thereby also generating limited adoption per Proposition 3.

We have noted that a novelty of our analysis is incorporating network delay into an economic model of blockchain. To emphasize the importance of incorporating this real-world feature that constitutes a practical constraint for Bitcoin, we provide the following result:

Proposition 5. No Adoption Problem Without Network Delay

Widespread adoption (i.e., $\lim_{N \rightarrow \infty} \pi_* > 0$) can be obtained within arbitrarily large networks (i.e., $\lim_{N \rightarrow \infty} M = \infty$) under the counterfactual assumption of no network delay (i.e., $\Delta(M) = 0$).
Propositions 4 and 5: consensus delay and zero-delay benchmark

for any reasonable model parameters. The implication of our findings is thus that Bitcoin is unlikely to replace traditional centralized payment systems.

4.1. Absolute bound on adoption level

There exists a bound on Bitcoin's adoption level that applies uniformly for all N :

Proposition 6. Finite Bound on Adoption Level

There exists a finite bound on the total adoption level that applies uniformly across all levels of transaction demand. (i.e., $\sup_{N \in \mathbb{N}} \pi_* N < \infty$).

Proposition 6 arises because an increase in the adoption level, $\pi_* N$, implies an increase in the endogenous fee level for the marginal user, $\phi(c_*)$. In turn, any adoption level beyond a particular finite bound cannot arise in equilibrium because any such adoption level implies a fee level that is too excessive to be consistent with such adoption levels.

More formally, Proposition 1 (C) implies the following:

$$R_1 \cdot \underbrace{\left(\frac{c_* - c}{c - c}\right)^\alpha \cdot N^\alpha}_{\text{Utility From Bitcoin}} + R_0 \geq \underbrace{\frac{N-1}{c - c} \times \frac{c_*^2 - c^2}{2\Lambda}}_{\text{Marginal User Fee}} \quad (9)$$

which states that the endogenous utility from using Bitcoin must exceed the fee paid for the marginal user. Then, using Eq. (2) and applying some algebra yields:

$$R_1 + \frac{R_0}{(\pi_* N)^\alpha} \geq \frac{N-1}{N} \times \frac{c_* + c}{2\Lambda} \times (\pi_* N)^{1-\alpha} \quad (10)$$

which reveals that arbitrarily large adoption levels cannot arise in equilibrium. In particular, letting the adoption level approach infinity in the limit as $N \rightarrow \infty$ in Eq. (10) yields:

Proposition 6: finite adoption bound

the case that the transaction rate varies with transaction demand:

Proposition 7. Finite Bound on Adoption Level II

Suppose that Bitcoin's transaction rate, Δ_N , varies with transaction demand, N . Then, even in such a case, Bitcoin faces a finite bound on its adoption level that applies uniformly across all levels of transaction demand. (i.e., $\sup_{N \in \mathbb{N}} \pi_* N < \infty$).

Proposition 7 arises for a similar reason as Proposition 3. In particular, Bitcoin faces a dilemma in the face of heightened transaction demand (i.e., when N is large). Either Bitcoin keeps transaction rates too low such that the congestion from fees ensures a finite bound on its adoption level (as per Proposition 6) or Bitcoin increases its transaction rate sufficiently fast such that wait times become prohibitive due to the need for consensus, also generating a finite bound on its adoption level. More formally, Proposition 1 (C) implies:

$$\underbrace{R_1 \cdot \left(\frac{c_* - \underline{c}}{\bar{c} - \underline{c}} \right)^\alpha}_{\text{Utility From Bitcoin}} \cdot N^\alpha + R_0 \geq \underbrace{c_* \Psi(\Delta_N, \Delta(M))}_{\text{Marginal Wait Disutility}} \quad (11)$$

which states that the endogenous utility from using Bitcoin must exceed the disutility from waiting incurred by the marginal user. Then, applying Eq. (2) and $\Psi(\Delta_N, \Delta(M)) \equiv \frac{1}{\Delta_N} + \tau(\Delta_N, \Delta(M)) \geq \tau(\Delta_N, \Delta(M))$ to Eq. (11) yields:

$$R_1 \cdot (\pi_* N)^\alpha + R_0 \geq c_* \tau(\Delta_N, \Delta(M)) \quad (12)$$

and dividing through by $(\pi_* N)^\alpha$ implies:

$$R_0 \leq c_* \tau(\Delta_N, \Delta(M))$$

8.5 Practical limits and potential solutions

Read:

- Proposition 8 translates the theory into practical limits for a VISA-scale transaction volume.
- Proposition 9 shows that partial centralization can improve adoption by reducing delay.
- Proposition 10 shows that adoption can be overcome if the protocol bounds fork probability away from one.

Economic message: Practical scalability requires more than bigger blocks or faster blocks. It requires protocol features that prevent fork probabilities from exploding.

hibitive fees implies that $\frac{c_{\text{tr}}(\Delta_N, \Delta(M))}{(\pi_* N)^\alpha} \rightarrow \infty$ as the adoption level grows arbitrarily large (i.e., $\pi_* N \rightarrow \infty$). Consequently, if the transaction rate grows sufficiently fast to avoid prohibitive fees, Eq. (13) implies $R_1 \geq \infty$, which is a contradiction and thereby precludes arbitrarily large adoption levels. Alternatively, if the transaction rate does not grow sufficiently fast to overcome prohibitive fees, then the level of the fees themselves preclude arbitrarily large adoption levels akin to the intuition given earlier for Proposition 6.

4.2. Practical adoption limits

We provide context regarding our previous analysis with the following result:

Proposition 8. Practical Limits of Adoption

Suppose that Bitcoin were to employ a transaction rate sufficient to process 150 million transactions per day and that not prohibit limited adoption (see Propositions 2 and 3), it does enable Bitcoin to achieve higher adoption rates for any finite level of transaction demand. We formalize this point with the following result:

Proposition 9. Partial Centralization of Mining Markets

Let π_*^C denote the adoption rate of Bitcoin if the network delay function is given by $\Delta_C(M)$. Let π_*^D denote the adoption rate of Bitcoin if the network delay function is given by $\Delta_D(M)$. We assume that $\Delta_C(M) \leq \Delta_D(M)$ for all M so that the former case reflects a (relatively) centralized mining market in which information propagates more quickly. Then, the adoption rate for the relatively centralized mining market exceeds that of the adoption rate for the relatively decentralized mining market (i.e., $\pi_*^C \geq \pi_*^D$).

Recall that M refers to the number of computer processors irrespective of the mining market structure. Since partial centralization of mining entails many such processors being controlled by the same entity, our model reflects partial centralization through a reduction in the network delay function (i.e., $\Delta_C(M) \leq \Delta_D(M)$ for all M). In that context, Proposition 9 demonstrates that a relatively

centralized mining market can achieve higher adoption rates than a decentralized mining market. We emphasize that a particular model-implied waiting time computation is not as relevant as the order of magnitude for wait times being unreasonably large whenever Bitcoin's transaction rate is expedited to tolerate large transaction demands.

To provide some practical context regarding the implications of persistent forks upon Bitcoin adoption, we turn to the only well-known example of a persistent Bitcoin fork. That fork demonstrates that persistent forks are a significant threat to widespread adoption in practice just as our model predicts. To provide more detail, a fork arose on the Bitcoin blockchain in March 2013 for technical reasons, but that fork was not resolved quickly so that it was persistent in a similar sense that forks generated in our model due to high transaction rates are persistent. In response to the fork, the largest cryptocurrency exchange at the time, Mt. Gox, suspended Bitcoin transactions, inducing a decline in Bitcoin usage. While Mt. Gox eventually

5.2.1. An expedited transaction rate

Several proposals aim to improve upon Bitcoin by employing a design identical to Bitcoin except with an expedited transaction rate. Such proposals are typically implemented either by increasing block sizes (e.g., Bitcoin Cash) or by increasing block rates (e.g., Litecoin). Our analysis indicates that such proposals do not succeed at overcoming limited adoption. In particular, Appendix A establishes that the fork probability approaches unity as the transaction rate diverges irrespective of whether the transaction rate diverges due to an increase in the block rate or an increase in the block size. Consequently, the consensus wait time becomes prohibitive as the transaction rate diverges irrespective of the implementation method for the increase in the transaction rate. In turn, expediting the transaction rate does not resolve the limited adoption problem as per Proposition 3.

While increasing the blockchain's transaction rate does not resolve the limited adoption problem, Proposition 10 indicates that limited adoption may be overcome by refining the blockchain's design to enable a high transaction rate without a high fork probability

9 Short summary

- The paper studies Bitcoin as a decentralized payment system with consensus and network delay.
- Users choose between Bitcoin and a traditional outside option.
- Users differ in waiting disutility.
- Miners enter freely when expected rewards and fees justify mining costs.
- More transaction demand raises fees.
- Higher fees attract more miners.
- More miners raise network delay.
- Network delay raises fork probability and consensus wait time.
- Longer consensus wait times reduce user adoption.
- With a fixed transaction rate, Bitcoin's adoption rate goes to zero as transaction demand grows.
- With a variable transaction rate, consensus wait time diverges when the transaction rate grows too fast.
- Therefore Bitcoin faces limited adoption both with fixed throughput and flexible throughput.
- There is a finite bound on total adoption for all transaction-demand levels.

- Matching traditional payment-system volume would imply extremely high fork probabilities and impractically long settlement times.
- Partial centralization can improve adoption, but it does not fully remove the bound.
- Alternative blockchain designs can overcome limited adoption if fork probability is bounded away from one.

10 Conclusion

10.1 Core conclusion

Hinzen, John, and Saleh show that Bitcoin’s limited adoption as a payment system can arise as an equilibrium outcome. The mechanism is not simply that Bitcoin has low transaction capacity. Rather, Bitcoin’s consensus architecture links transaction demand, fees, miner entry, network delay, and forks in a way that creates negative network effects.

10.2 Economic intuition

A centralized payment system can usually scale by adding processing capacity. Bitcoin cannot do so in the same way because faster processing increases fork risk. Forks delay consensus, and consensus is required for transaction settlement. Thus, the very features that make Bitcoin decentralized also limit its ability to scale as a payment system.

10.3 Takeaway

The paper’s main lesson is that decentralization has economic costs. Bitcoin’s design requires consensus among many miners, and consensus becomes harder as the network expands and transaction rates rise. Widespread adoption requires not only more throughput but a protocol that limits consensus failures and settlement delays.